



National University of Science and Technology
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INTERPRETABLE MACHINE LEARNING FOR URBAN HOUSING AFFORDABILITY

**A DATA-DRIVEN ANALYSIS OF U.S.
METROPOLITAN MARKETS**

MASTER'S THESIS DEFENSE



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RESEARCH CONTEXT & MOTIVATION

Urban Housing Affordability and Sustainability Challenges



Motivation	Problem
Sustainable development requires balancing economic, social, and environmental dimensions. SDG 11 connects housing affordability with social equity and urban resilience.	Broad sustainability frameworks are difficult to translate into measurable, data-driven analysis. This thesis narrows the focus to U.S. metropolitan housing markets, where rising prices, rental pressure, and regional disparities can be examined through measurable indicators.



RESEARCH AIM & QUESTIONS



Aim: To explain housing price dynamics using visual analytics and interpretable tree-based machine learning

RQ1 What are the primary drivers of housing price variation across U.S. metropolitan regions?

RQ2 Can tree-based machine learning models outperform linear regression?

RQ3 How do housing prices and affordability metrics evolve across U.S. regions between 2018 and 2025?

RQ4 What interpretable patterns can be extracted from tree-based models?

LITERATURE REVIEW & CONTRIBUTION



Literature Review Across Three Domains



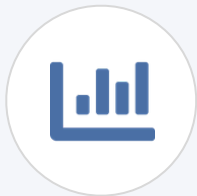
Sustainability & SDGs

SDG 11 frames housing as a measurable urban sustainability challenge.



Urban Housing and Affordability

Affordability, rental pressure, and regional disparities shape housing market dynamics.



Machine Learning Approaches in Housing Analysis

Tree-based models capture nonlinear price patterns while supporting interpretation.

PRISMA SYSTEMATIC REVIEW

247

Identified



246

Screened



109

Eligible



20

Included

CONTRIBUTION OF THIS THESIS

Integrates SDG 11 framing, Zillow housing indicators, data analytics and tree-based machine learning to explain U.S. metropolitan housing price dynamics.

DATA PREPROCESSING & CLEANING

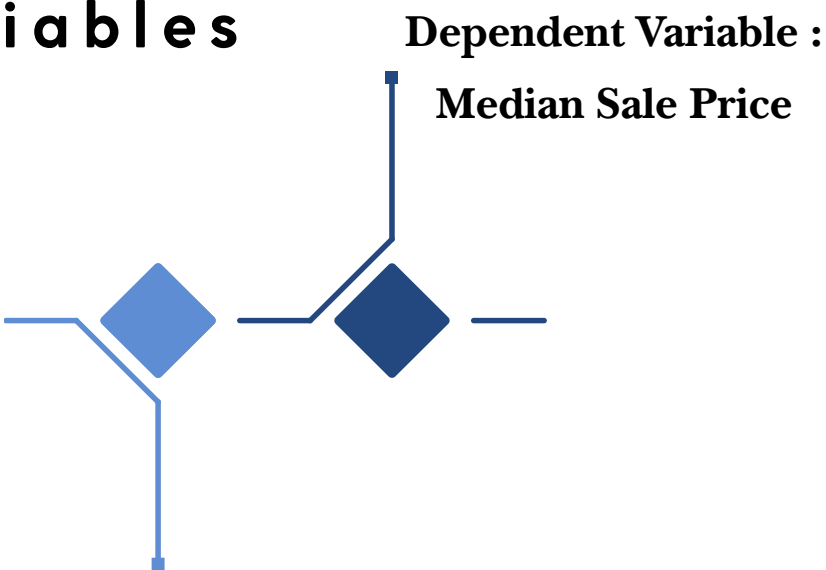


Dataset Overview

SOURCE	TIME SPAN
Zillow Research	Mar 2018 – Feb 2025
ROWS × COLS	GEOGRAPHY
9,893 × 12	U.S. MSAs

7 Zillow datasets merged by RegionID + Date

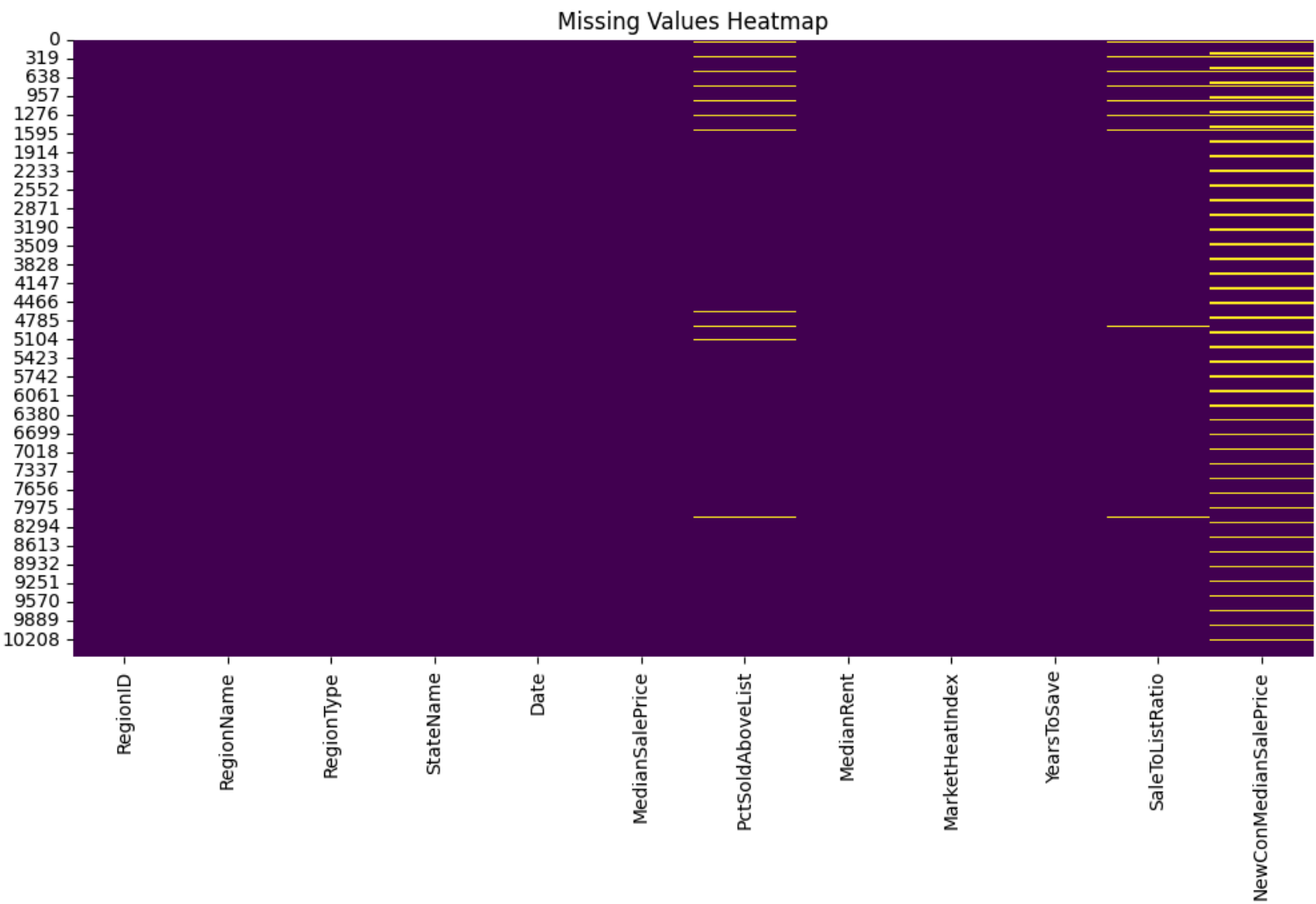
Key Variables



Independent Variables:

- Years To Save · Median Rent (Affordability Metrics)
- Pct Sold Above List · Sale To List Ratio · Market Heat Index (Market Competitiveness)
- New Con Median Sale Price (Supply-Side Indicator)

10500 ROWS → 9893 ROWS



Rows before drop: 10500
Rows after drop (excluding only key variables): 9893

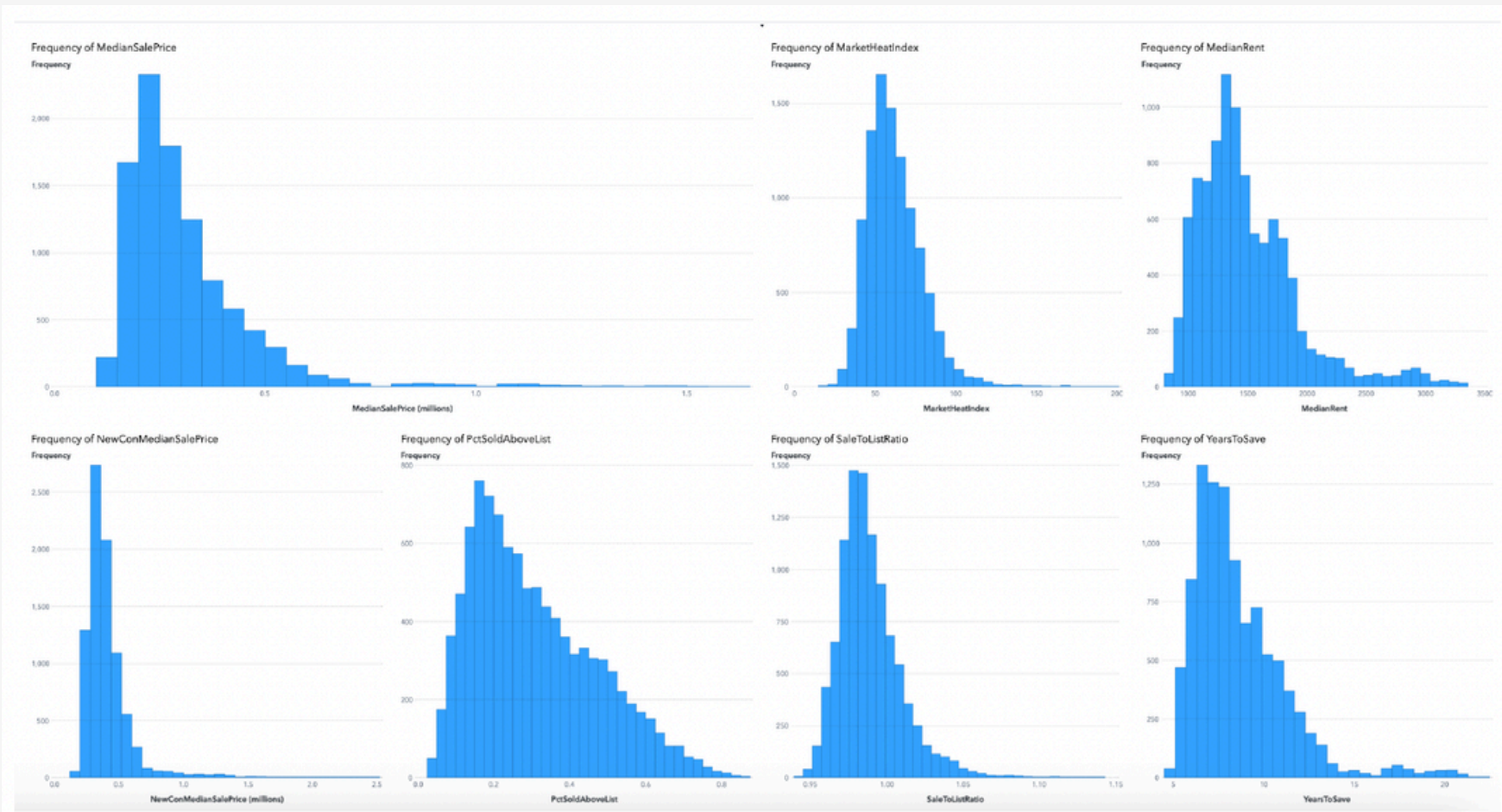
Duplicate Rows: 0

Not dropped missing values in NewconMedian SalePrice

EXPLORATORY HOUSING MARKET ANALYSIS

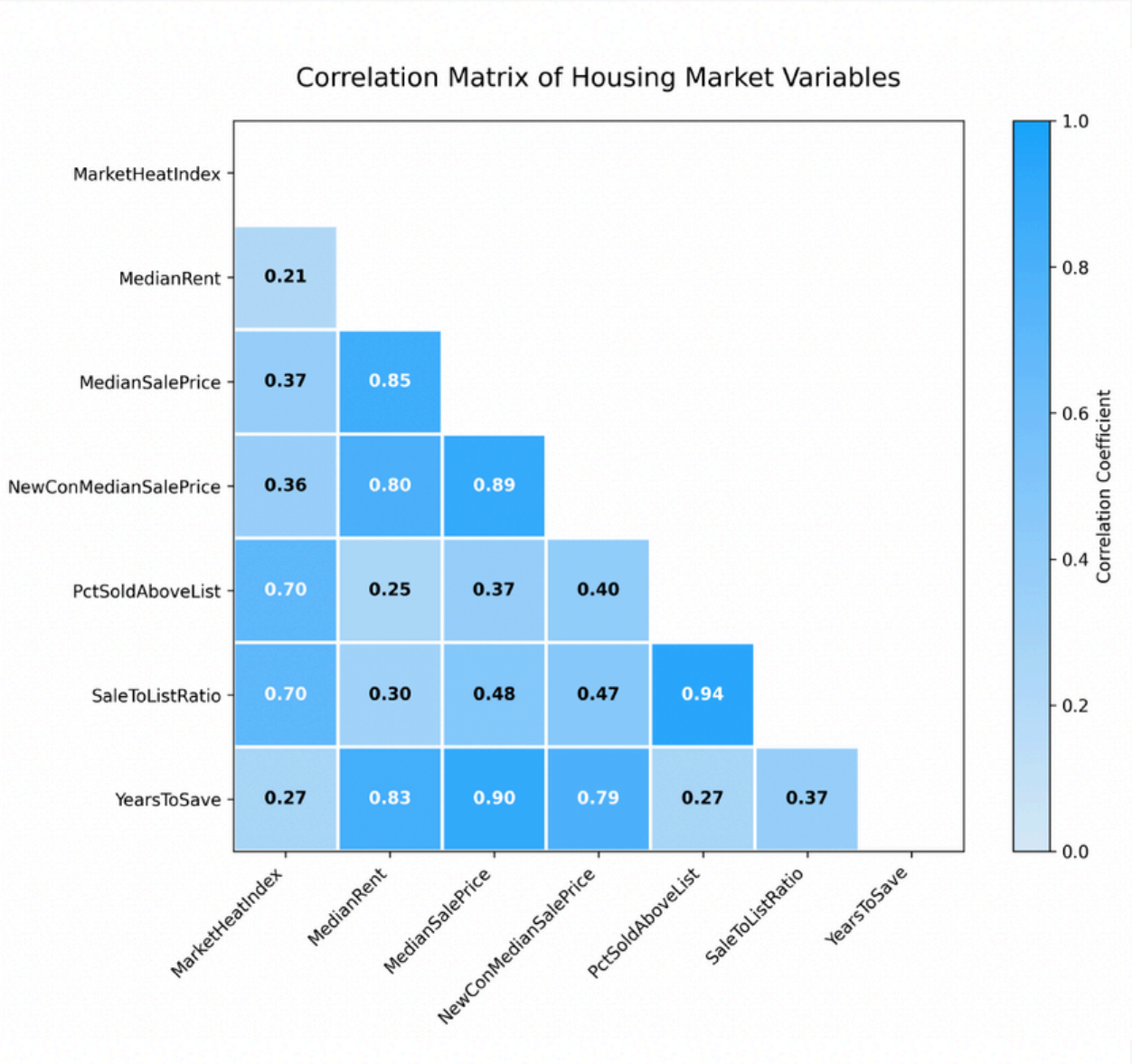
Distributions & Relationships Analysis

Distribution Analysis



Right-skewed distributions indicate **heterogeneous** and **nonlinear** housing markets

Correlation Matrix



$r = 0.90$ MedianSalePrice $\leftrightarrow$ YearsToSave

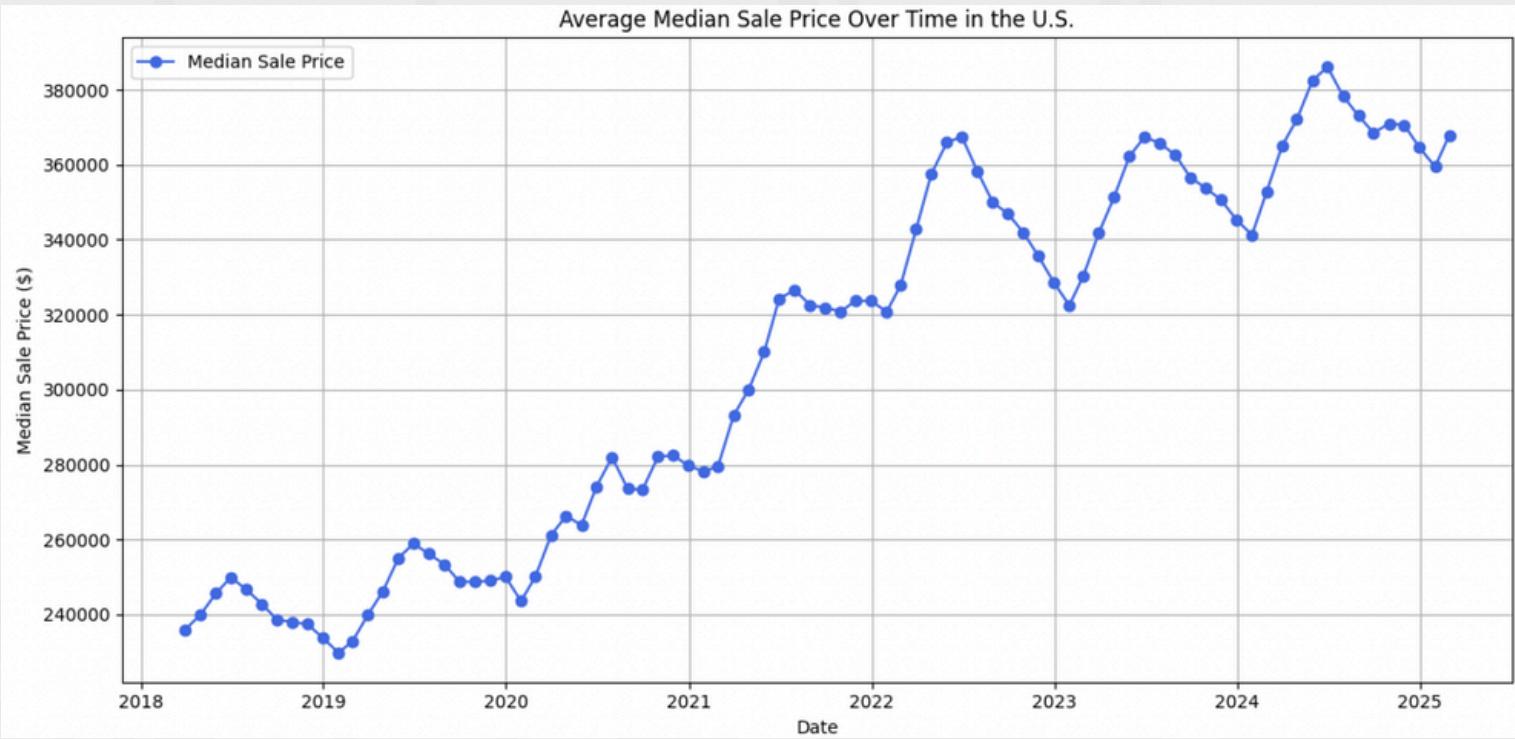
$r = 0.85$ MedianSalePrice $\leftrightarrow$ MedianRent

Affordability indicators ($r \geq 0.83$) outperform market competitiveness variables ($r \leq 0.37$) in explaining price variation.

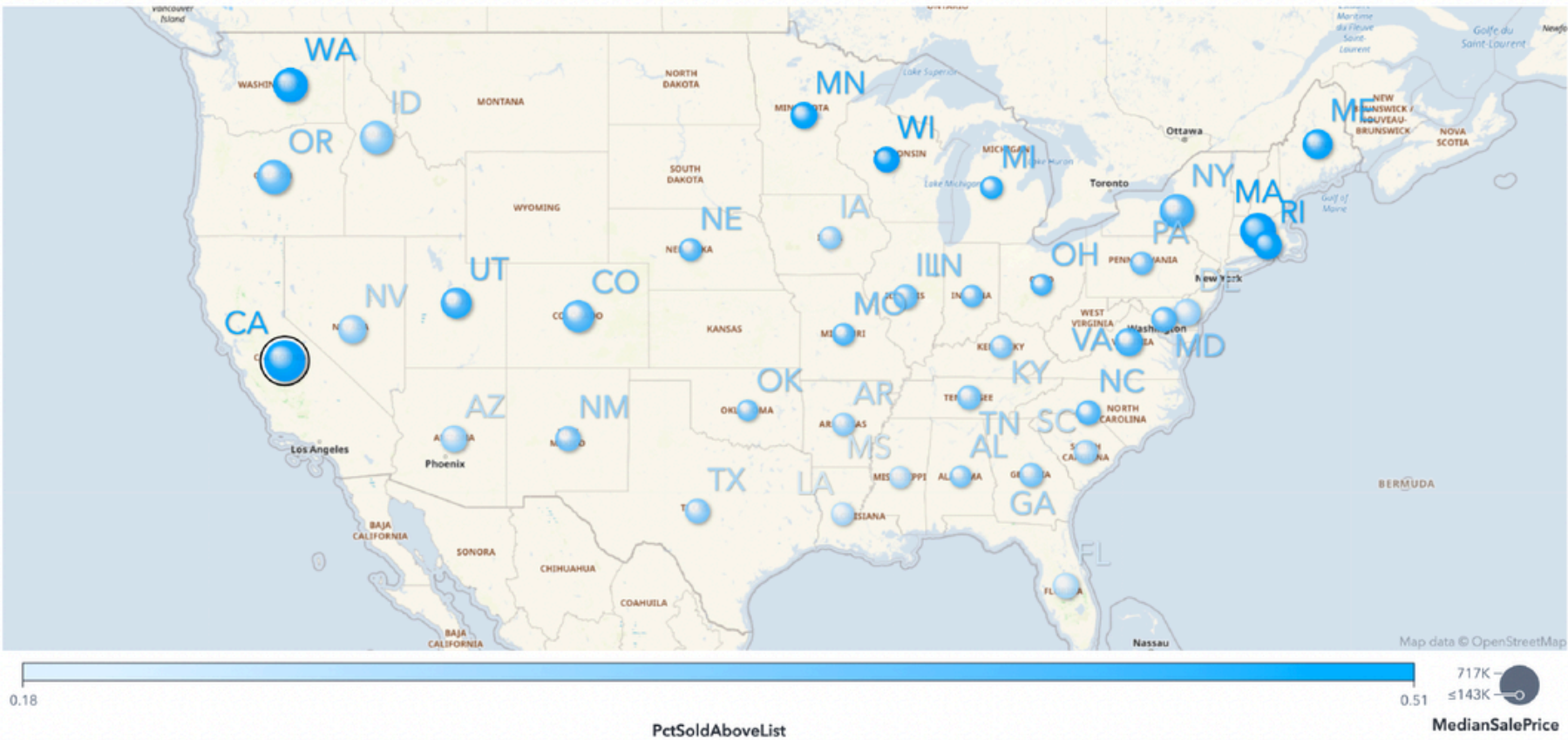
Temporal & Regional Analysis

How Housing Pressure Evolves Over Time and Across Regions

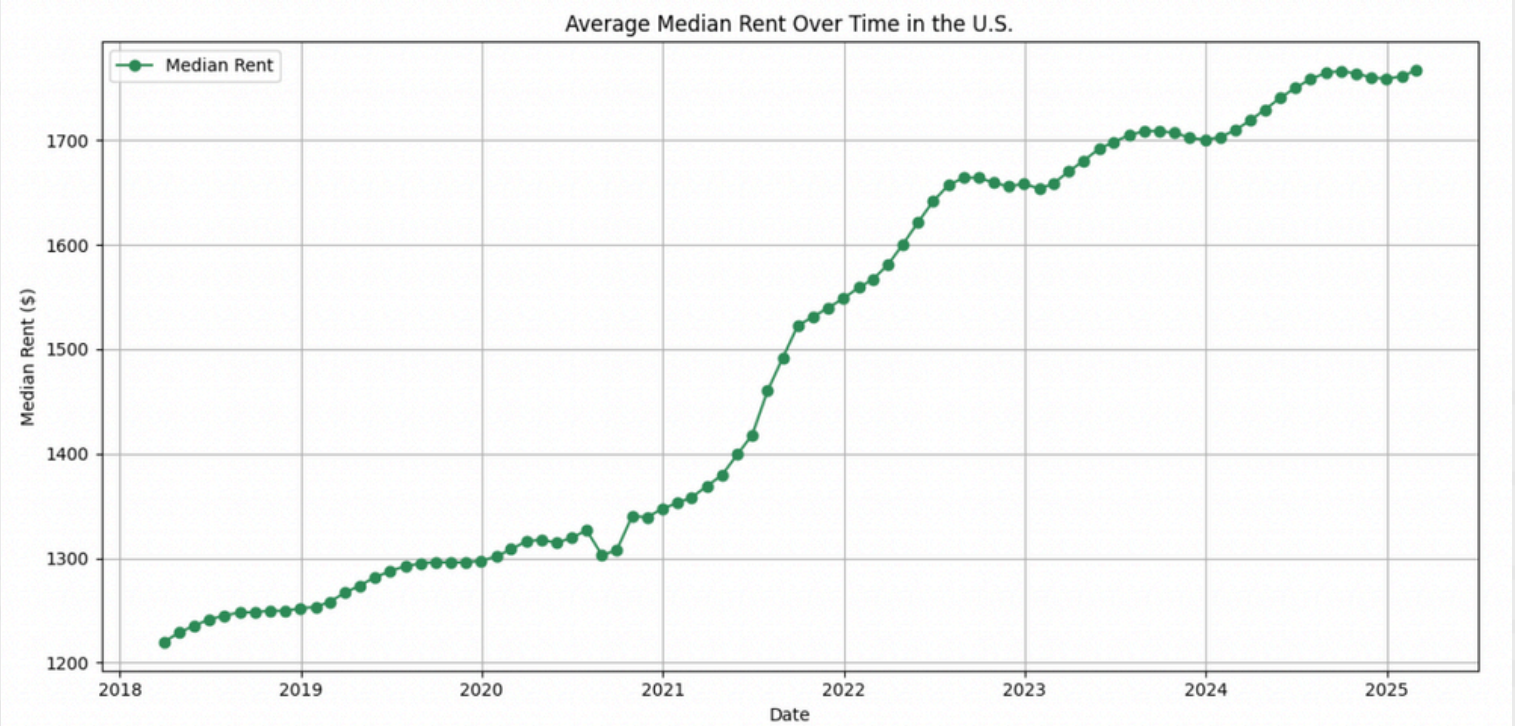
Median Sale Price (2018–2025)



Regional Distribution



Median Rent (2018–2025)



Geo Bubble Map

Bubble size = Median Sale Price Color = % Sold Above List

Findings

- Sharpest increase **2021–2022**, followed by fluctuations at a higher level.
- **CA** shows the **highest median sale price**, while **MA** and **WI** show the **strongest market competition**.
- High prices concentrate in coastal markets, but strong competition also emerges in moderate-price regions.

MODELLING FRAMEWORK



A Two-Phase Approach: Preliminary Selection and Tree-Based Evaluation

PHASE 1

Preliminary Model Selection

Four candidate models were evaluated using SAS Visual Analytics.

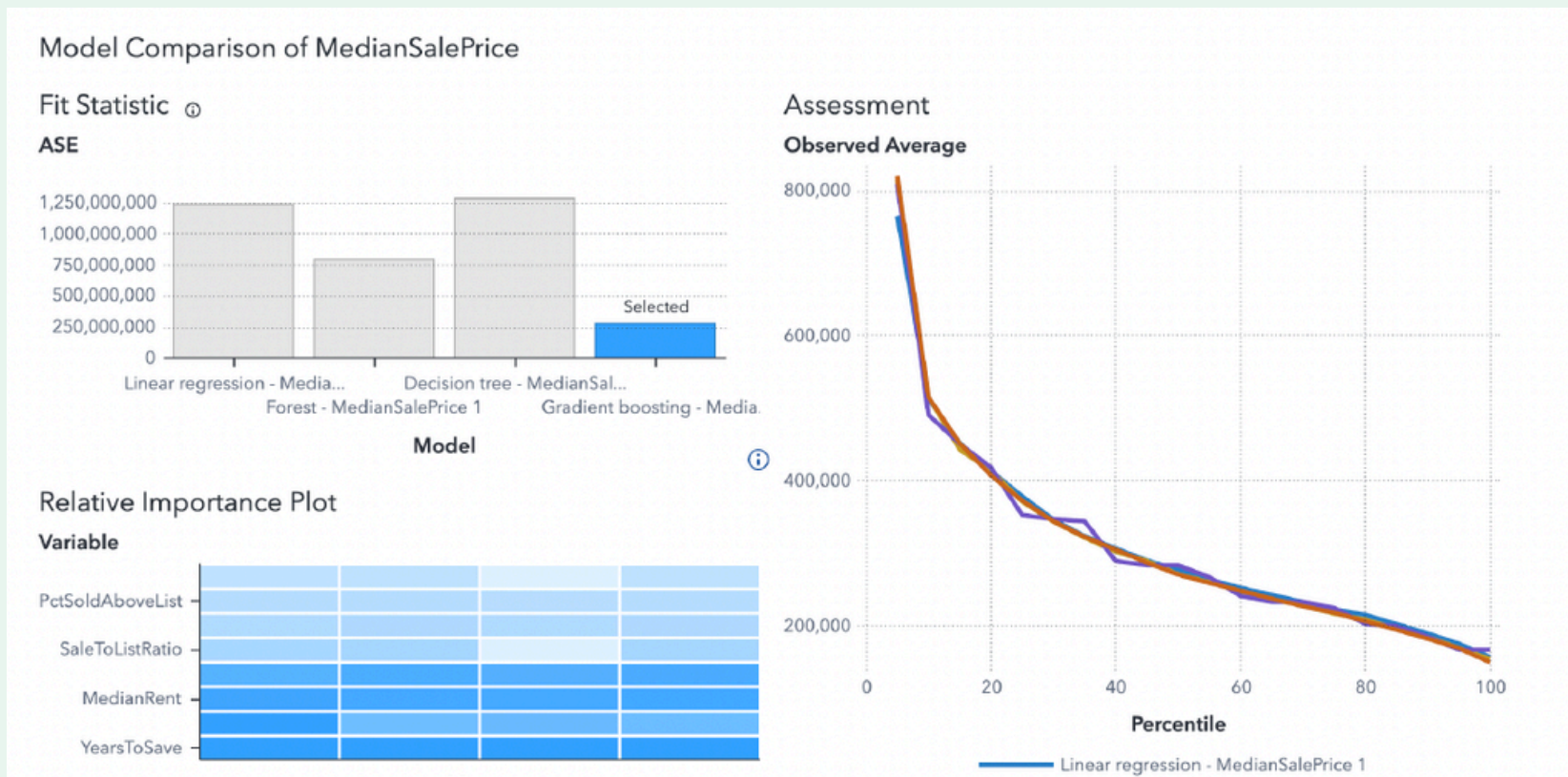


Fig. 3.15 — ASE comparison, variable importance, assessment curves

Finding: Ensemble models outperformed the linear baseline. **YearsToSave** and **MedianRent** ranked as the strongest predictors.

PHASE 2

Tree-Based Modelling and Evaluation

Shared setup, all models:

Target: Median Sale Price

Split: **60%** Train / **20%** Validation / **20%** Test

Default and **tuned** configurations

Champion selected on test partition

Implementation via SAS Viya - Model Studio

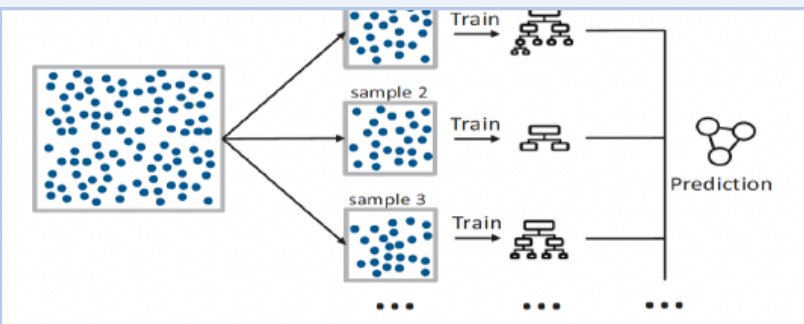
Models evaluated:

Decision Tree *Interpretable baseline*

Variance reduction splits
Pre-pruned using depth and leaf-size limits

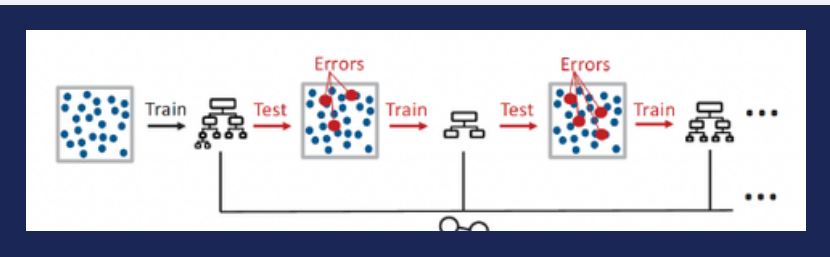
Random Forest

Bagging, variance reduction



Gradient Boosting

Boosting, bias reduction



All models were trained on identical partitions, ensuring a fair comparison across modelling strategies.

MODEL RESULTS & COMPARISON

Predictive performance on the test partition

Family	Config.	RMSE (\$)	MAE (\$)	R ²
Decision Tree	Default	48,339	32,159	0.918
Decision Tree	Tuned	38,999	25,416	0.947
Random Forest	Default	28,869	17,538	0.971
Random Forest	Tuned	33,918	20,444	0.960
Gradient Boosting	Default	29,616	19,008	0.969
Gradient Boosting	Tuned	19,461	12,911	0.987

- Ensembles sharply reduce test error.
- Gradient Boosting captures nonlinear price patterns best.

Best Performing Models

1. Tuned Gradient Boosting

2. Default Random Forest

3. Default Gradient Boosting

Ensemble models outperform the single-tree baseline, with **tuned Gradient Boosting** as the **champion**.

CHAMPION MODEL RESULTS

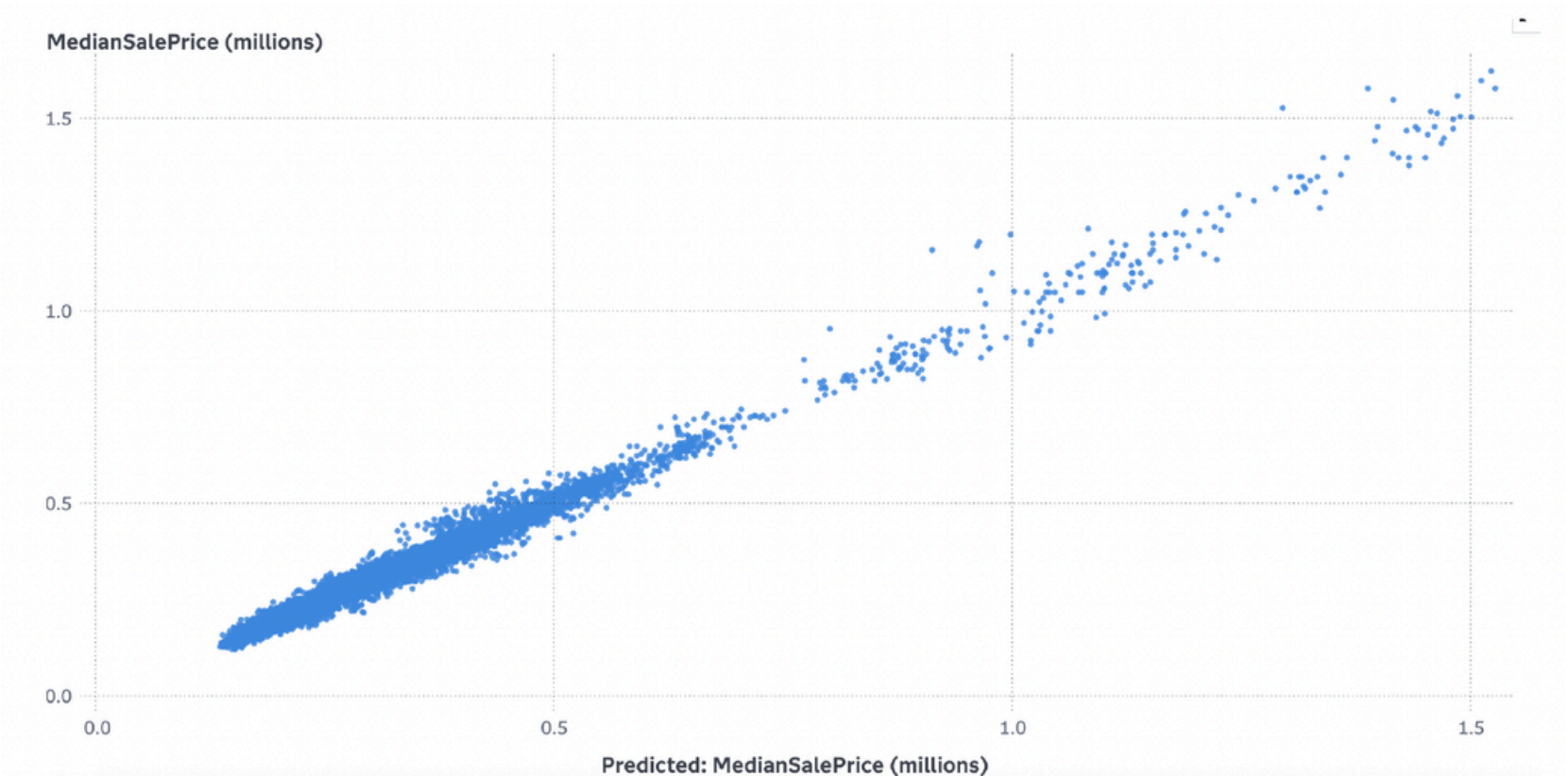


FIT STATISTICS ACROSS PARTITIONS FOR CHAMPION MODEL

Partition	N	RMSE (\$)	MAE (\$)	R ²
Train	5,935	11,946	8,820	0.995
Validation	1,979	19,733	12,959	0.985
Test	1,979	19,461	12,911	0.987

Train, validation and test scores remain close, $R^2 = 0.995 / 0.985 / 0.987$, indicating stable generalisation.

PREDICTED VS. ACTUAL (FULL TEST PARTITION)

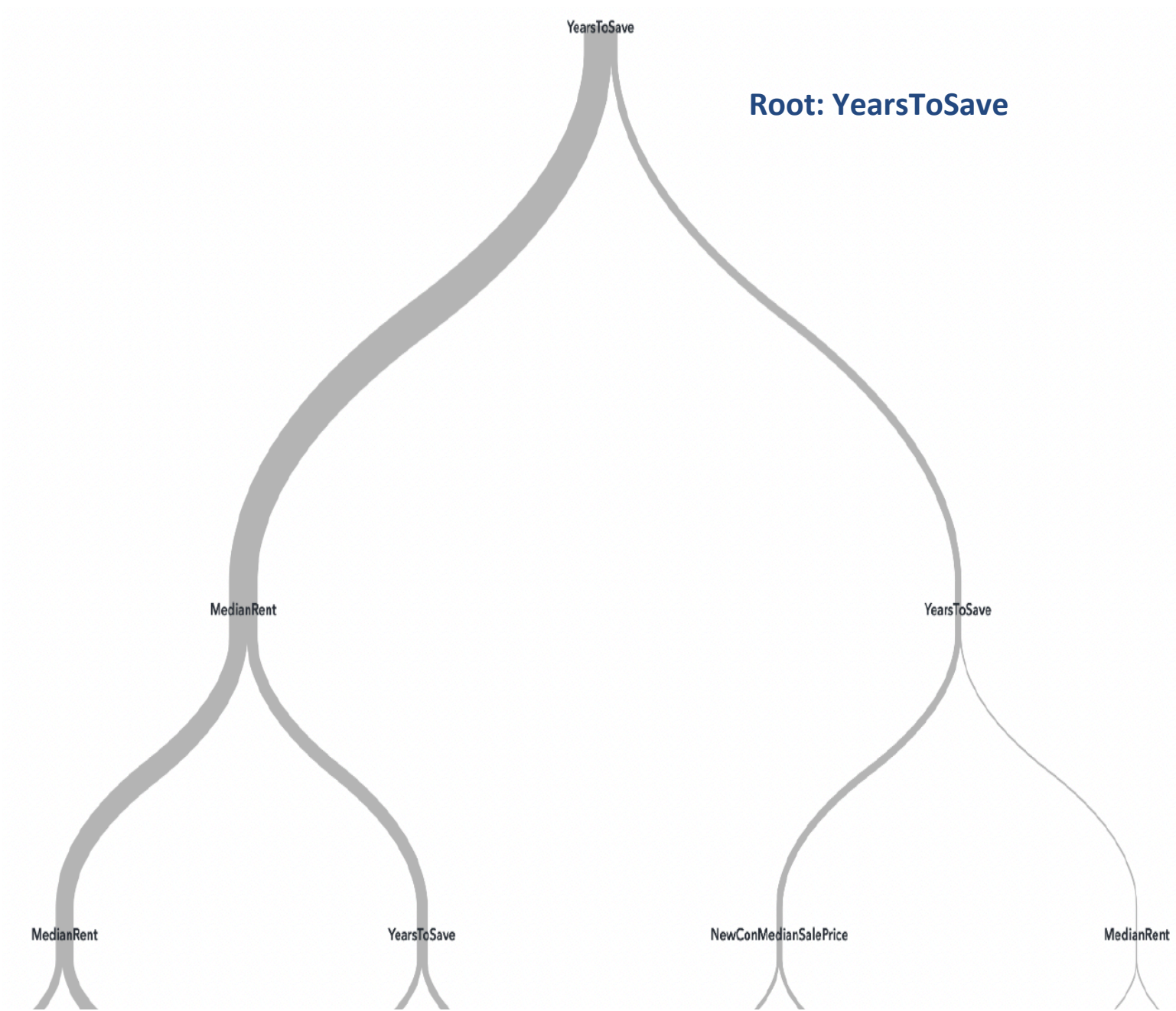


Predictions closely follow actual prices across the full range.

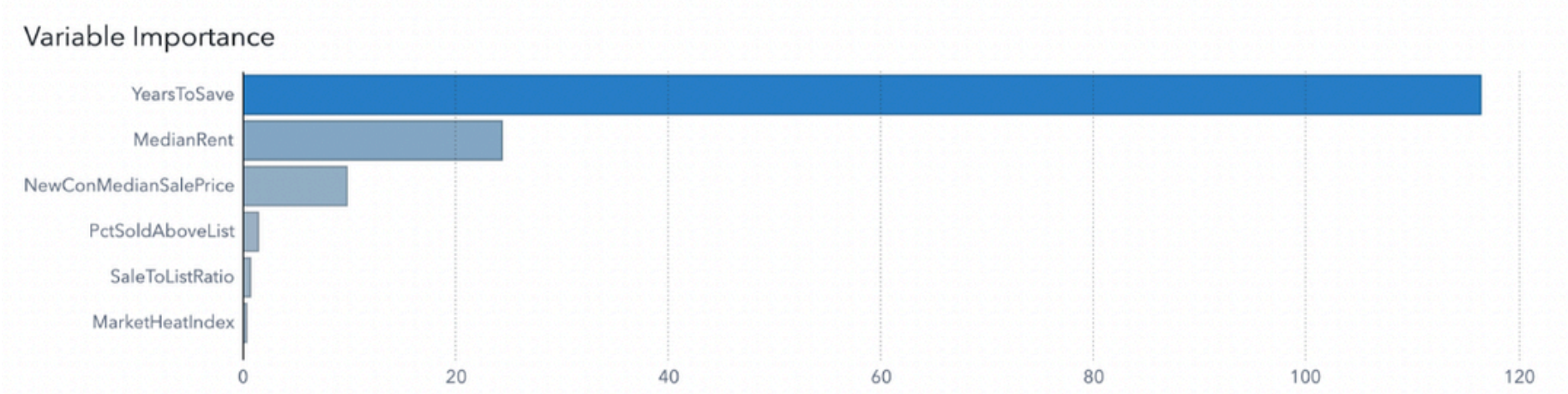
Champion model: Accurate on unseen data and stable across partitions.

INTERPRETABILITY & KEY FINDINGS

Regression Tree Structure



Variable Importance



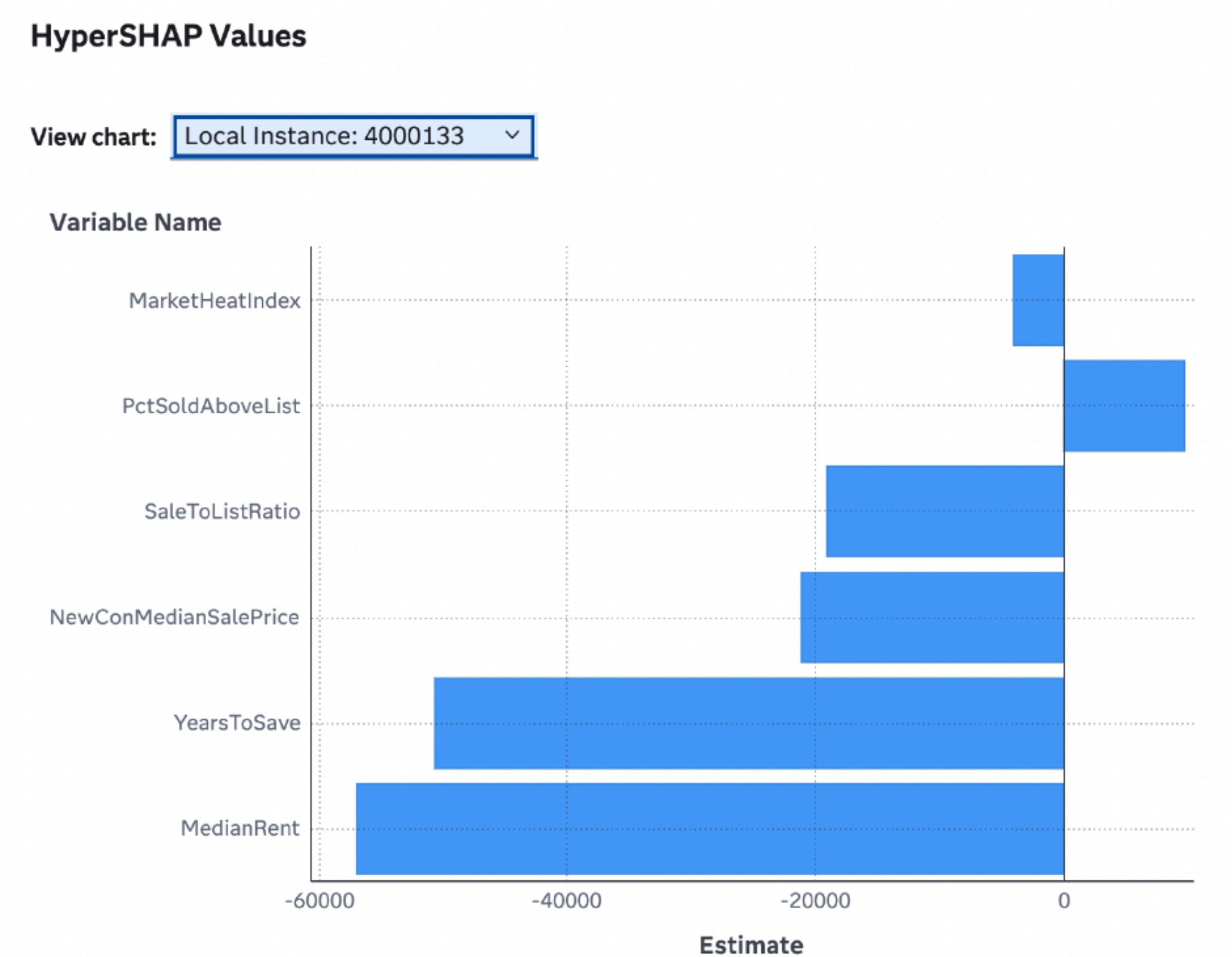
YearsToSave
Primary driver. First split across all tree models.
MedianRent Secondary driver.
Competitiveness variables Appear only in lower-level splits.

Housing price segmentation is driven by **long-term affordability, not short-term market competition.**

HYPERSHAP LOCAL EXPLANATIONS



Explains the direction and relative strength of each variable’s effect on one prediction.



Generated via SAS Model Studio's post-training interpretability framework, **applied to tuned Gradient Boosting champion model.**

- **MedianRent and YearsToSave -> strongest negative contributors.**
- **PctSoldAboveList → a small positive counter-effect.**

CONCLUSION



ANSWERING THE FOUR RESEARCH QUESTIONS

RQ1

Primary drivers of price

YearsToSave (#1) and MedianRent (#2) are the dominant predictors, market-competition variables ranked lower.

RQ2

Tree-Based Models vs. Linear Baseline

Tuned Gradient Boosting gives the best result, reducing test RMSE from \$48.3K to \$19.5K with $R^2 = 0.987$, outperforming the linear baseline.

RQ3

Temporal Trend: 2018–2025

Sharp increase in prices and rents in 2021–2022, followed by stabilisation at high levels.
High prices concentrate in coastal metropolitan regions.

RQ4

Interpretable patterns

Variable importance, icicle plots, and HyperSHAP independently converge on the same affordability-based segmentation.

LIMITATIONS & FUTURE WORK



LIMITATIONS



Single data source

All indicators from Zillow; U.S. metros only, may not transfer to rural or smaller markets.



Limited feature set

Interest rates, household income and housing supply were not included.



Static models

Observations treated independently; temporal structure is not modelled.



Association, not causation

Models reveal which variables relate to price, not what causes price changes.

FUTURE RESEARCH DIRECTIONS



Broader geographies

Apply the framework to international cities and varied economic conditions.



Richer features

Add interest rates, income, supply and sustainability indicators (transport, inequality).



Temporal modelling

Use time-series methods to capture how housing markets evolve over time.



Deeper analysis

Analyse causal methods, also partial dependence & accumulated local effects.

THANK YOU